

Airframe noise sources and reduction technologies in aircraft

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Abstract

The continuous increase of the air traffic linked to the operations around airports leads to increase the community noise problem. Aircraft noise is an issue of enormous environmental and technological impact. Projected growth in significant quieting of modern jet engine has brought renewed attention to the nonpropulsive component of aircraft noise. Airframe-generated noise can be the dominant component of the total noise radiated from commercial aircraft, particularly for large aircraft and especially during the landing approach. In this paper, a review of major components of airframe noise and technologies with the greatest potential to reduce airframe noise are presented.

Keywords: Aircraft noise; Airframe noise; Aircraft noise mitigation; Airframe noise reduction; noise reduction technologies.

1. Introduction

Communities' adjacent to major airports are often exposed to high noise levels from airframe-generated sound from commercial airliners. Land-use planning has in a few cases enabled new airports to be built far from the centers of population. However, soon after an airport has been built, housing developments emerge and encroach on the airport; inevitably, new residents will make claims that the noise from civil aircraft is intolerable. The control of aircraft noise in the vicinity of airports from the late 1960s to the present day has been achieved by setting statutory noise limits and monitoring the aircraft noise [1, 2]. Aircraft main noise sources are the engines, the airframe, and the interference between the engines and the airframe. During landing, when the engines are operating at reduced power, the noise from the airframe, of which landing gear noise is an important part, is equal to the engine noise. During take-off and flyover, when the engines develop maximum power, the engines are the dominant noise

source. The airframe noise would be the noise produced by the aircraft, if all engines were made inoperative [3]. The dominant sources of airframe noise are known to be associated with unsteadiness of the separated and/or vertical flow regions around the high-lift system (i.e., flaps and slats) and the aircraft undercarriage (i.e., landing gear).

While a clean airframe produces a broadband source, strong tones may be observed due to the wing trailing edge vortex shedding. Low-frequency tones have also been identified in association with cavities [4] or gaps [5] in the airframe.

In this paper, we present an overview of the major noise generation mechanisms associated with an airframe and its reduction concepts.

2. Airframe noise

Airframe noise is the main contributor to overall noise during landing approach of an aircraft when it is equipped with quiet high bypass ratio engines [6]. The success of the high bypass ratio turbofan engine in reducing the external noise of civil transport aircraft at take-off and landing, while improving the economics of air travel, has opened up the debate as to how much further it will be possible to reduce aircraft noise by the introduction of new aircraft or existing aircraft retrofitted with new engines [7].

Therefore, airframe noise is the noise generated by an aircraft flying through the air without the propulsion system or auxiliary power units operating. This noise is produced by air flowing over and around the aircraft. It becomes most serious for an aircraft in the landing configuration, where regions of turbulent flow are produced when the wing slats and flaps are extended and the landing gear are lowered.

Airframe noise sources are the result of the interaction of flow fluctuations with the surface (hard walls) and there especially with surface irregularities (rapid change of boundary conditions). The character of the generated noise sources, mean directivity and spectral distribution of sound pressure is strongly influenced by the character of the

flow (boundary layer status, Reynolds number, Strouhal number, Mach number). The following flow types when interacting with hard walls are of main interest (Figure. 1):

- Turbulent boundary layers passing sharp trailing edges
- Laminar or transitional boundary layers coming along with periodic variation of the mean flow by instabilities passing sharp trailing edges
- Separated flows passing sharp trailing edges
- Attached flows passing blunt trailing edges
- Flow around side edges (development of side edge vortices).

Further noise sources may be generated by gaps or cavities like fuel overpressure holes [8].

Figure 2 from Reference [9] illustrates some of the airframe noise sources, which include the fuselage, main wing, landing gear, and high-lift devices, etc.

Even though each of these noise sources generates sound by different physical mechanisms, the noise generated is similar in amplitude but can have different noise directivity patterns. The noise from the flaps

and slats is characterized by turbulent boundary layers passing over sharp trailing edges and side edges.

3. Landing gear

The landing gear is an inevitable system for the aircraft. It absorbs the energy of the landing impact and carries the aircraft weight at all ground operations, including take off, taxiing, and towing. Noise generated by aircraft landing gear is a major contributor to the overall airframe noise during landing approach of a commercial aircraft [10-13]. The landing gear represents a significant challenge because of the non-uniform flow field on the wing and complex landing gear geometries with small-scale details that affect the high frequency noise. Interactions between the landing gear wake and the flap system contribute to the complicated noise signature [14]. The noise source is the turbulent, unsteady, separated flow around various components of the landing gear. Since the landing gear has many cavities and sharp edges, the flow-field is very complex with three-dimensional separation regions of different sizes [15]. Most of the remainder of this noise is a result of bluff body separation and shedding from the various components. Because the

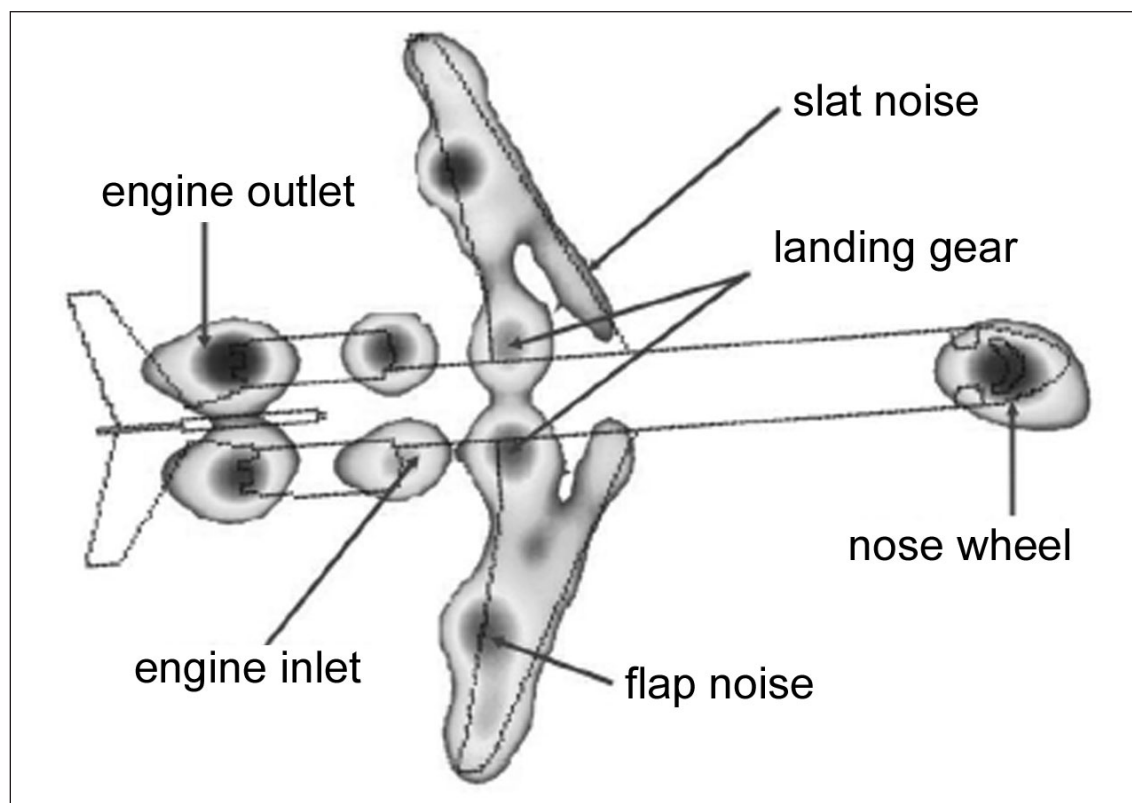


Figure 1. Some of the airframe noise sources [9]

size of the components varies greatly, the noise spectrum is broadband. However, the peak in the spectrum appears to be fairly low (around a few hundred hertz). Removal of the small parts such as hydraulic lines and nuts has been shown to reduce the high frequency noise by several decibels. Although actual removal of the small parts would be impractical, some of the small components could be hidden behind larger ones in relatively dead zones of the flow. Some small components could also be enclosed by the larger ones [1].

Landing gear noise intensity increases proportionally to the 6th power of flow velocity while the sound frequencies increase linearly with the velocity [16].

The reduction of bluff gear components and the passive control of vortex shedding via splitter plates behind the cylindrical struts are the solutions commonly adopted to reduce the landing-gear noise [17]. Furthermore, using fairings to shield components from high speed impingement reduces noise. The effect of bleeding air through a fairing on aero acoustic performance of landing gear is a relatively new topic. Recent research on the subject [18] has confirmed the flight test results, although the corresponding flow behavior has not been investigated thoroughly.

A general theory that describes a passive control method for bluff body aerodynamic noise claimed that if two bodies can be treated as acoustically compact and are placed in close proximity, then noise cancellation will occur if favorable amplitude and phase relationships can be achieved [18].

Li and his colleagues [19] studied the effectiveness of using plasma actuators to control broadband noise generated by a bluff body flow experimentally. They have studied the potential of plasma actuators to reduce landing gear noise during approach phase of an aircraft. The control effectiveness of both dielectric barrier discharge and sliding discharge plasma actuators were tested in laboratory environment, using a representative bluff body consisting of a circular cylinder and an oblique strut. They have showed that the flow speed impinging on the downstream strut was reduced by the upstream plasma forcing, contributing to the reduced noise.

4. High lift device

High lift devices include leading edge slats and trailing edge flaps. A slat, when deployed, forms a cove region between the slat and the central main element of the aircraft wing. Flow separation, flow recirculation, an unsteady shear layer, and slat settings together

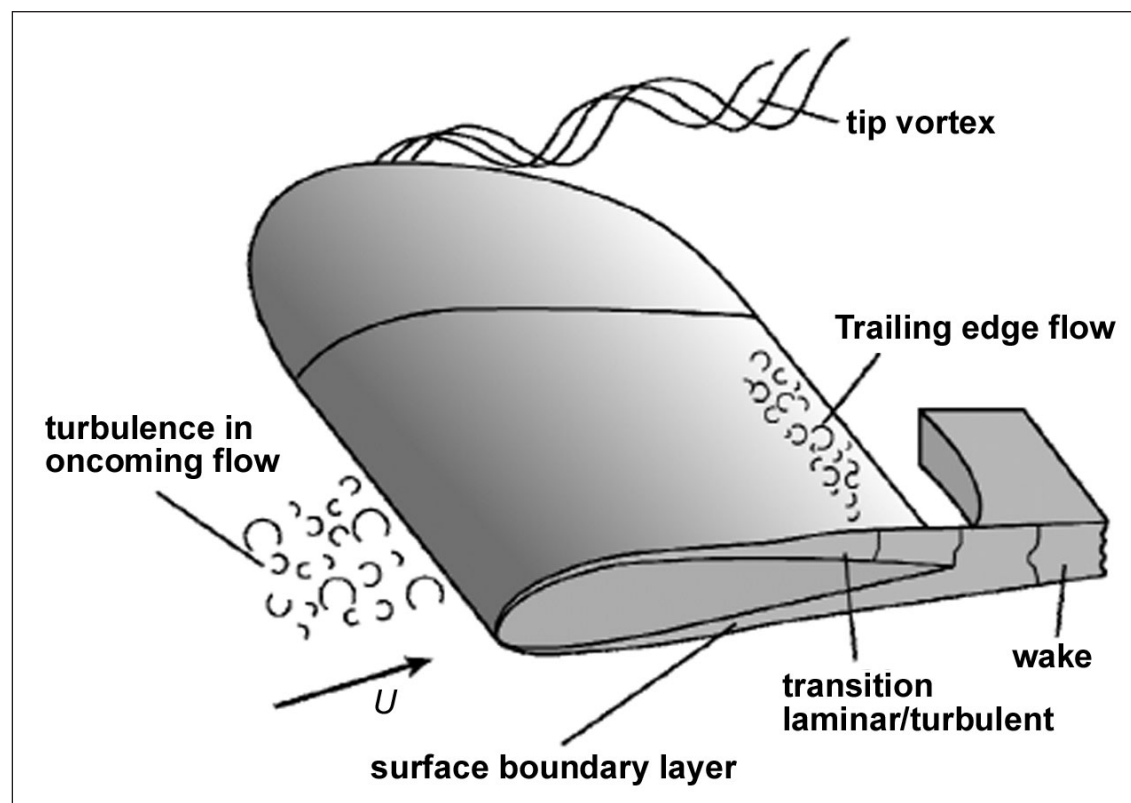


Figure 2. Sources of noise on a typical wing [9]

generate noise of mainly broadband content. For a flap, the outboard flap side edge and vortex system associated with it are the main sources of noise. The intensity of high lift device noise generally follows a power law of flow velocity [20]. High lift devices noise characteristics were investigated both numerically [21, 22] and experimentally [23, 24] at model and full scale. However, due to the extremely complex configuration and the aerodynamics associated with wing high lift devices, the development of reliable noise prediction schemes and effective noise reduction devices remains in a research stage.

4.1. Slat

Due to the finite thickness of the slat trailing edge, the boundary layers on the pressure and suction sides of the slat separate near the trailing edge and form free shear layers that roll up and shed strong spanwise vortices in the form of a vortex street. Interaction of the vortex street with the trailing-edge surface produces high-amplitude pressure oscillations that radiate noise to the far field. For a noise reduction technique to be effective, it must either eliminate or significantly diminish the vortex shedding and, in so doing, reduce the intensity of the pressure fluctuations near the slat trailing edge [25].

M. R. Khorrami, et al. [26] used a Porous trailing-edge treatment as a passive means for slat noise reduction. They showed that Assessment of the pressure fluctuations in the vicinity of the treated edge indicates a potential noise reduction in excess of 20 dB. The primary mechanism underlying this reduction is related to the reduced strength of Strouhal shedding from the finite thickness trailing edge. A secondary effect of the treatment involves an upward shift in the Strouhal-shedding frequency to a frequency band of reduced auditory sensitivity in a full-scale application.

Experiments have shown that brush-like devices installed on slat and flap tips reduce the noise levels significantly. An extension of this method is the installation of brushes on the entire slat trailing edge to reduce its noise [27, 28].

While the character of radiated noise from the airframe is generally broadband in nature, strong tones have been observed to occur in both experiments and full-scale flyover tests. The presence of these tones is highly undesirable due to their effect in significantly increasing the perceived noise level. The origin of many of these tones has been found to be

the region around the leading edge slat and trailing edge flap [29].

4.2. Trailing edge noise

Trailing edge noise sources, like those from the flaps or slats, have been studied extensively [30, 31]. TE noise is directly related to the turbulence produced in the boundary layer of an airfoil. When vortices pass the TE, sound is emitted due to the abrupt change in boundary condition.

The airframe high-lift noise, intuitively thinking, is configuration dependent. Any increase in the deflection angle of leading edge slat and trailing edge flap will lead to an increase in overall noise level. Larger their deflection angles are, the higher the noise level will be. This is due to the fact that the high-lift components are more loaded when their deflection angles are increased, hence loading noise; in addition, higher deflection may induce stronger vortex shedding and turbulent wake downstream which causes higher noise as a result [32].

4.3. Flap side edge noise

Flap side edge flows have long been recognized as a significant contributor to airframe noise. The noise source mechanisms are related to the vortex structure of the cross flows in the flap side edge region. Many authors have studied flap side edge noise [33-41].

The flap edge vortex is always a strong source of the airframe noise, especially when high lift devices are fully deployed during take-off or landing. According to Prandtl's classic lifting-line theory, a trailing vortex will be generated whenever there is a change in the bound circulation over the wing. For a wing flap configuration, the lift and hence the bound circulation is much higher over the flap than on the main wing. Thus the circulation will not be continuous at the interface between the wing and the flap, and a very strong vortex will be generated here. These vortices have been seen in many experiments and flight tests. For example, the experimental [42] and computational [43] studies indicated that a very strong vortex was generated at the flap-edge due to the sudden increase in the lift. This vortex, due to its interaction with the flap gap, will generate a strong noise, commonly labeled as "flap edge noise".

In theory, the source intensity should vary according to the fifth or sixth power of aircraft speed. Experimental studies, involving the identification and separation the various

noise sources for analysis, seem to indicate that the velocity dependence may be of lower power dependence [44].

Experimental data are reviewed to show that fences can modify the cross flow in the side edge region of aircraft flaps in such a way that noise reduction can be achieved. Near field pressure data reveal that fences reduce the characteristic frequency of the flow fluctuations. Flap side edge fences are simple devices; they are basically flat plates attached to the side surfaces of the flaps in such a way that the plates protrude into the flow on the lower side of the flaps. It has been confirmed in wind tunnel tests that the fences only change the local flows in that the overall lift characteristics of the flaps and the high lift systems is not affected by the fences in any significant way.

For noise exclusively from flap side edges, fences can achieve a reduction of about 4 dB for the EPNL, but the reduction in EPNL dB can be expected to be much less for the total airframe noise. This is especially true for cases where other sources, such as those related to the slats and landing gears, are equally or more important but basically unaffected by flap fences. This is indeed the case for modern commercial airplanes, which tend to have fewer flap elements and lower flap deployment angles, because of advanced high lift system design. This reduces the relative importance of flap-related noise sources, and thus, reduces the benefit of any flap noise reduction devices, including the fences [45].

Micro-tabs are a row of trapezoidal fingers, slightly higher than the boundary layer thickness, attached to the pressure side of the flap close to its edge.

In study in University of California by Kuo and Sarigul-Klijn [46], numerical methods were carried to study the two dimensional unsteady flow field and acoustic components around a high-lift airfoil. Micro-tab device was designed and attached to the pressure side of the flap near its trailing-edge such that settings of high-lift devices could be reduced and hence the related airframe noise level was lowered. They report that with the deployment of the micro-tab device, noise content in the mid-frequency range was reduced by 3–5 dB on the average as compared to the baseline configuration. No significant acoustic penalty was observed when comparing the micro-tab case with the other reduced setting case without the tab deployment, except the low-frequency tone below 100 Hz, which was then speculated to

be due to drag oscillation in the flow field around the tabbed airfoil. Three parametric studies were then performed: the tab location study showed that although the Gurney flap configuration (placing tab right at the flap trailing-edge) was more efficient in generating lift, it also produced notable noise in particular a tone in the mid-frequency range. By varying the angle of attack on the tabbed airfoil, it was shown that increasing AOA might have helped reduce slat-related noise, which was consistent with the previous research findings.

In another study, Kuo and Sarigul-Klijn [32] reported that through the reduced high-lift settings and the micro-tab application, an overall 2.3 dB noise reduction was achieved. They showed that the low frequency tone was totally eliminated by the slit, along with a broadband noise level increase in the high frequency domain due to stronger downstream wake instability. An overall noise level decrease of 2.4 dB was observed.

But a few issues still require further investigation. For instance, onset of the spanwise instability which usually results in rollup from the side-edge of the obstacle (i.e. micro-tab) was not able to be distinguished, and thus might require improvement in the configuration setup for the high-lift wing. However, micro-tab proposed by the authors demonstrated its potential to be applied onto commercial airliners for the use in airframe noise reduction during approach to landing phase of flight.

The possibility of reducing noise further is to use trailing edge brushes [47]. Finez, et. Al. [48] used a brush made of a single row of flexible polypropylene fibers is integrated in the trailing edge of a cambered airfoil. Far field acoustic measurements show a noise reduction potential reaching 3 dB on a wide frequency range. Due to high curvature of the incident flow, a secondary acoustic source partly masks the trailing edge noise reduction.

5. Conclusions

With the efficient reduction of engine noise, airframe noise has emerged as an ever more important component of the overall aircraft noise especially during aircraft approach and landing when the engines are at low power. The main noise sources of airframe can be classified as Landing gear noise and High lift devices noise containing flap edge side noise, flap trailing edge noise and slat leading edge noise. In this paper we discussed the main technologies for airframe noise reduction such

as fence, micro-tab, trailing edge brushes and Porous trailing-edge treatment.

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One complaint stops the clock

A Croxley Green church which has had its 140-year-old hourly chimes silenced following a single complaint from a resident has launched a legal challenge against Three Rivers District Council. All Saints Church in The Green was served with a noise abatement notice, banning chimes between 11pm and 7am, on September 26 after environmental health officers upheld a complaint from a local resident. Because it is not possible to set the ancient mechanism to work within specific hours, the bells have been silenced to comply with the order, but a legal challenge is already underway. Reverend Miriam Mugan revealed this week that the number of letters and emails from residents supporting the bells made the decision to instruct diocesan solicitors to appeal against the order through the magistrates court easier. She said: "The clock was installed in 1877 and the mechanism cannot be turned off for a few hours or on for a few hours. We have turned it off because we need to abide by the terms of a noise abatement notice. This has been a tough decision for us because of the financial expense we will incur. However, we feel we cannot allow this decision to go unchallenged. We do not know the timescale yet, it could be months rather than weeks, but until then we must abide by the noise abatement notice." Scores of concerned residents crowded into the community centre on Community Way at a recent extraordinary meeting of the parish council which saw councillors pledge their unanimous support for All Saints Church. During the meeting, the audience reacted angrily to the news that the same resident previously made a complaint about the clock chimes in 2004, which was dismissed by the council. Councillor Philip Brading, who is also the district council's member for environmental health, said: "I had a discussion with the environmental health officer charged with investigating this complaint and this isn't something the officer has done lightly or on a whim. A complaint was received on August 10 the first thing they did was call the legal department for advice and the chief executive of Three Rivers. They then carried out a night time visit. She reached the conclusion that in that house, in that room, the noise was such that it would constitute a statutory noise nuisance, having determined that she was under obligation to act."